

Sangeek Hyun

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RESEARCH SUMMARY

I am a Ph.D. candidate in Artificial Intelligence at Sungkyunkwan University, working at the intersection of generative modeling, computer vision, and visual representation learning. My research aims to build scalable generative systems that can model, synthesize, and manipulate visual worlds across images, videos, and 3D scenes. Recently, I am particularly interested in developing visual world models that learn from real-world scenes and simulate spatially and temporally coherent visual environments based on my background knowledge in scalable generative modeling and video/3D generation.

RESEARCH INTERESTS

- Generative Models, Visual World Model, Video and 3D Generation, 3D Scene Representation, Efficient and Scalable Generation.

EDUCATION

Sungkyunkwan University (SKKU) Ph.D. Candidate, Artificial Intelligence (Visual Computing Lab) • Advisor: Prof. Jae-Pil Heo	Suwon, South Korea <i>Sep. 2021 – Present</i>
Sungkyunkwan University (SKKU) M.S. in Artificial Intelligence (Visual Computing Lab) • Advisor: Prof. Jae-Pil Heo	Suwon, South Korea <i>Feb. 2020 – Aug. 2021</i>
Sungkyunkwan University (SKKU) B.S. in Computer Science and Engineering	Suwon, South Korea <i>Mar. 2016 – Feb. 2020</i>

HONORS AND AWARDS

- **Winner**, Qualcomm Innovation Fellowship Korea (QIFK) 2024

PUBLICATIONS

Summary: 7 first-/co-first-author papers and 11 co-authored papers at top-tier conferences and journals including CVPR, NeurIPS, ICLR, ICML, ICCV and ECCV. My work has received **933** citations (May 30, 2026, Google Scholar).

Cross-scale Aligned Supervision for Training GANs

Sangeek Hyun, MinKyu Lee, Jae-Pil Heo

arXiv preprint 2026

Disambiguating 2D-3D Correspondences in Gaussian Splatting-based Feature Fields for Visual Localization

Miso Lee, Sangeek Hyun, Yerim Jeon, Jae-Pil Heo

arXiv preprint 2026

Scalable GANs with Transformers

Sangeek Hyun, MinKyu Lee, Jae-Pil Heo

International Conference on Machine Learning (ICML) 2026

SeaCache: Spectral-Evolution-Aware Cache for Accelerating Diffusion Models

Jiwoo Chung, Sangeek Hyun, MinKyu Lee, Byeongju Han, Geonho Cha, Dongyoon Wee, Youngjun Hong, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) **[Oral]** 2026

Looking Beyond the Window: Global-Local Aligned CLIP for Training-free Open-Vocabulary Semantic Segmentation	
ByeongCheol Lee, Hyun Seok Seong, Sangeek Hyun , Gilhan Park, WonJun Moon, and Jae-Pil Heo	
<i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>	2026
Rethinking LayerNorm in Image Restoration Transformers	
MinKyu Lee, Sangeek Hyun , Woojin Jun, Hyunjun Kim, Jiwoo Chung, Jae-Pil Heo	
<i>International Conference on Learning Representations (ICLR)</i>	2026
Correlation-guided calibration of query dependency for video temporal grounding	
WonJun Moon, Sangeek Hyun , SuBeen Lee, Jae-Pil Heo	
<i>Pattern Recognition PR</i>	2025
Diffusion Feature Field for Text-based 3D Editing with Gaussian Splatting	
Eunseo Koh, Sangeek Hyun , MinKyu Lee, Jiwoo Chung, Kangmin Seo, Jae-Pil Heo	
<i>Advances in Neural Information Processing Systems (NeurIPS)</i>	2025
Fine-Tuning Visual Autoregressive Models for Subject-Driven Generation	
Jiwoo Chung, Sangeek Hyun , Hyunjun Kim, Eunseo Koh, MinKyu Lee, Jae-Pil Heo	
<i>International Conference on Computer Vision (ICCV)</i>	2025
AESOP: Auto-Encoded Supervision for Perceptual Image Super-Resolution	
MinKyu Lee, Sangeek Hyun , Woojin Jun, Jae-Pil Heo	
<i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>	2025
GSGAN: Adversarial Learning for Hierarchical Generation of 3D Gaussian Splats	
Sangeek Hyun , Jae-Pil Heo	
<i>Advances in Neural Information Processing Systems (NeurIPS)</i>	2024
Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer	
Jiwoo Chung*, Sangeek Hyun* , Jae-Pil Heo (*Equal contribution)	
<i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i> [Highlight]	2024
Diversity-aware Channel Pruning for StyleGAN Compression	
Jiwoo Chung, Sangeek Hyun , Sang-Heon Shim, Jae-Pil Heo	
<i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>	2024
Task-disruptive Background Suppression for Few-Shot Segmentation	
Suho Park, SuBeen Lee, Sangeek Hyun , Hyun Seok Seong, Jae-Pil Heo	
<i>AAAI Conference on Artificial Intelligence (AAAI)</i>	2024
Frequency-based Motion Representation for Video Generative Adversarial Networks	
Sangeek Hyun , Jaihyun Lew, Jiwoo Chung, Euiyeon Kim, Jae-Pil Heo	
<i>IEEE Transactions on Image Processing (TIP)</i>	2023
Disentangled Representation Learning for Unsupervised Neural Quantization	
Haechan Noh, Sangeek Hyun , Woojin Jeong, Hanshin Lim, Jae-Pil Heo	
<i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>	2023
Query-dependent Video Representation for Moment Retrieval and Highlight Detection	
WonJun Moon*, Sangeek Hyun* , SangUk Park, Dongchan Park, Jae-Pil Heo (*Equal contribution)	
<i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>	2023

Local Attention Pyramid for Scene Image Generation

Sang-Heon Shim, Sangeek Hyun, DaeHyun Bae, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2022

Self-Supervised Video GANs: Learning for Appearance Consistency and Motion Coherency

Sangeek Hyun, Jihwan Kim, Jae-Pil Heo

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

2021

VarSR: Variational Super-Resolution Network for Very Low Resolution Images

Sangeek Hyun, Jae-Pil Heo

European Conference on Computer Vision (ECCV)

2020

ACADEMIC SERVICES

Conference Reviewer: CVPR (2022–2026), ICCV (2025), ECCV (2024, 2026), NeurIPS (2024–2026), ICLR (2025–2026), ICML (2025–2026), AAAI (2026)

Journal Reviewer: IJCV (2025–2026)

PROJECTS

Named Entity Recognition for Video Understanding

Apr. 2025 – Mar. 2026

- Developed video NER models for entity extraction and video understanding using multimodal large language models, and built evaluation pipelines for systematic benchmarking.

Detection of AI-Generated Visual Content

Jul. 2021 – Feb. 2025

- Co-led a project on AI-generated visual content detection, including dataset construction and the development of deepfake and synthetic media detection models. * *Recognized by the Minister of Science and ICT for outstanding achievement in social problem-solving R&D.*

Fashion Image Retrieval and Generation

Sep. 2020 – Dec. 2021

- Developed models for fashion image retrieval and generation, focusing on image synthesis and visual feature learning for fashion applications.

Handwritten Mathematical Expression Recognition

Mar. 2020 – Dec. 2020

- Developed handwritten mathematical expression recognition models and built paired datasets of handwritten formula images and LaTeX annotations.

Semiconductor Defect Detection and Classification

Sep. 2019 – Feb. 2020

- Developed computer vision models for semiconductor defect detection and classification with an emphasis on robust visual inspection in few-shot industrial data.